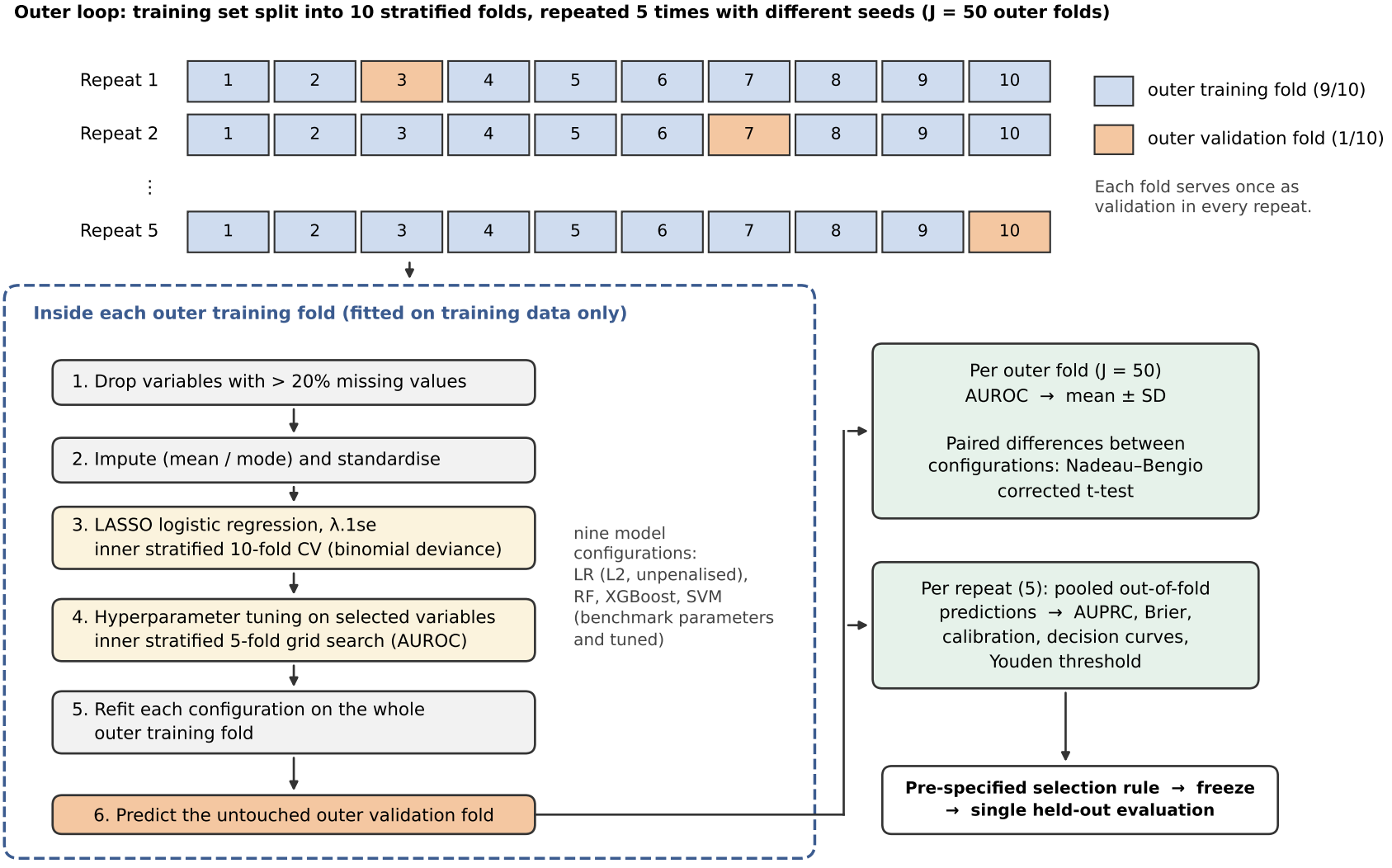
Outer loop: training set split into 10 stratified folds, repeated 5 times with different seeds (J = 50 outer folds)
Repeat 1
Repeat 2
Repeat 5
1 2 3 4 5 6 7 8 9 10
outer training fold (9/10)
outer validation fold (1/10)
Each fold serves once as validation in every repeat.
Inside each outer training fold (fitted on training data only)
1. Drop variables with > 20% missing values
2. Impute (mean / mode) and standardise
3. LASSO logistic regression, λ.1se
inner stratified 10-fold CV (binomial deviance)
4. Hyperparameter tuning on selected variables
inner stratified 5-fold grid search (AUROC)
5. Refit each configuration on the whole
outer training fold
6. Predict the untouched outer validation fold
nine model configurations:
LR (L2, unpenalised),
RF, XGBoost, SVM
(benchmark parameters and tuned)
Per outer fold (J = 50)
AUROC → mean ± SD
Paired differences between configurations: Nadeau–Bengio corrected t-test
Per repeat (5): pooled out-of-fold predictions → AUPRC, Brier, calibration, decision curves, Youden threshold
Pre-specified selection rule → freeze → single held-out evaluation